



friend function :

- friend function is not a member function of a class to which it is a **friend**.
 - friend function is declared in the class with **friend** keyword.
 - It must be defined outside the class to which it is friend.
 - friend function can access any member **{ private + public }** of the class to which it is friend.
 - friend function cannot access members of the class **directly**.
 - It has no caller object.
 - It should not be defined with **membership label**.
- ★ A friend function, is a function "that is" allowed to access the **private** and **protected** or **public** members of a class even though it is not a member of that class.



Program: #include <iostream>

class complex

{

private:

int a, b;

public:

void setData(int x, int y)

{

a = x;

b = y;

}

void showData()

{

cout << "a = " << a << "b = " << b;

}

};

friend void fun(complex); → friend function declaration

};

void fun(complex c) → friend function definition

{

cout << "sum is " << c.a + c.b;

}

void main()

{

complex c1, c2, c3;

c1.setData();

fun(c1);

}

Because friend function cannot directly access the class members

friend function:

friend function can become friend to more than one class.

Program: #include <iostream>

class A

{

private:

int a;

public:

void setData(int x)

{

a = x;

}

friend void fun(A, B);

}

class B

{

private:

int b;

public:

void setData(int y)

{

b = y;

}

friend void fun(A, B);

}

void fun(A a1, B a2)

{

cout << "sum is " << a1.a + a2.b; → friend function accessing private member from two different class at same time.

}



void main()

{

A obj1;

B obj2;

obj1.setData(2);

obj2.setData(3);

}

fun(obj1, obj2);

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friend function: Overloading of operators by using friend function

Program: #include <iostream>

class complex

{

private:

int a, b;

public:

void setData(int x, int y)

{

a = x;

b = y;

}

void showData()

{

cout << "a = " << a << "b = " << b;

}

friend complex operator+(Complex X, Complex Y);

};

Complex operator+(Complex X, Complex Y)

{

Complex temp;

temp.a = X.a + Y.a;

temp.b = X.b + Y.b;

return(temp);

}

void main()

{

Complex c1, c2, c3;

c1.setData(3, 4);

c2.setData(5, 6);

c3 = c1 + c2;

c3.showData();

}

→ c3 = operator+(c1, c2)

friend function: overloading of unary operator using friend function

Program: #include <iostream>

class complex

{

private:

int a, b;

public:

void setData (int x, int y)

{

a = x;

b = y;

}

void showData()

{

cout << "a = " << a << "b = " << b;

}

friend complex operator- (complex);

};

complex operator- (complex x)

{

complex temp;

temp.a = -x.a;

temp.b = -x.b;

return temp;

}

void main()

{

complex c1, c2;

c1.setData(3, 4);

c2 = -c1;

c1.showData();

c2.showData();

}

* Output =

a = 3 b = 4

a = -3 b = -4

→ c2 = operator-(c1)

3 + 17 = 20

friend function: overloading of Insertion and Extraction operators

Program: #include <iostream>

class complex

{

private:

int a, b;

public:

void setData(int x, int y)

{

a = x;

b = y;

}

void showData()

{

cout << "In a = " << a << "b = " << b;

}

friend ostream & operator << (ostream &, complex);

friend istream & operator >> (istream &, complex);

};

ostream & operator << (ostream & dout, complex c)

{

dout << "In a = " << c.a << "b = " << c.b;

return (dout);

}

istream & operator >> (istream & din, complex &c)

{

din >> c.a >> c.b;

return (din);

}



```
void main()
```

```
{
```

```
    Complex c1;
```

```
    cout << "Enter a complex number ";
```

```
    cin >> c1;
```

```
    cout << "You entered :";
```

```
    cout << c1;
```

```
}
```

Output :

Enter a complex number 4 5

You entered : 4 + 5i

a=4 b=5

01/09/2017 - 11/11

the end of the bridge to rockmen taking at least a whole day
to go round all the rockmen's settlements and return to the boat with

modem 220/17

$l = l + \text{right}$ → grünster
 $d = d + \text{right}$ → gallen für grüner
 $d = d + \text{right}$ → grün

1. $\frac{1}{2}$ of 100 = 50
 2. $\frac{1}{4}$ of 100 = 25
 3. $\frac{1}{8}$ of 100 = 12.5
 4. $\frac{1}{16}$ of 100 = 6.25
 5. $\frac{1}{32}$ of 100 = 3.125
 6. $\frac{1}{64}$ of 100 = 1.5625
 7. $\frac{1}{128}$ of 100 = 0.78125
 8. $\frac{1}{256}$ of 100 = 0.390625
 9. $\frac{1}{512}$ of 100 = 0.1953125
 10. $\frac{1}{1024}$ of 100 = 0.09765625
 11. $\frac{1}{2048}$ of 100 = 0.048828125
 12. $\frac{1}{4096}$ of 100 = 0.0244140625
 13. $\frac{1}{8192}$ of 100 = 0.01220703125
 14. $\frac{1}{16384}$ of 100 = 0.006103515625
 15. $\frac{1}{32768}$ of 100 = 0.0030517578125
 16. $\frac{1}{65536}$ of 100 = 0.00152587890625
 17. $\frac{1}{131072}$ of 100 = 0.000762939453125
 18. $\frac{1}{262144}$ of 100 = 0.0003814697265625
 19. $\frac{1}{524288}$ of 100 = 0.00019073486328125
 20. $\frac{1}{1048576}$ of 100 = 9.5367431640625e-05
 21. $\frac{1}{2097152}$ of 100 = 4.76837158203125e-05
 22. $\frac{1}{4194304}$ of 100 = 2.384185791015625e-05
 23. $\frac{1}{8388608}$ of 100 = 1.1920928955078125e-05
 24. $\frac{1}{16777216}$ of 100 = 5.9604644775390625e-06
 25. $\frac{1}{33554432}$ of 100 = 2.98023223876953125e-06
 26. $\frac{1}{67108864}$ of 100 = 1.4901161193847656e-06
 27. $\frac{1}{134217728}$ of 100 = 7.450580596923828e-07
 28. $\frac{1}{268435456}$ of 100 = 3.725290298461914e-07
 29. $\frac{1}{536870912}$ of 100 = 1.862645149230957e-07
 30. $\frac{1}{1073741824}$ of 100 = 9.313225746154785e-08
 31. $\frac{1}{2147483648}$ of 100 = 4.656612873077392e-08
 32. $\frac{1}{4294967296}$ of 100 = 2.328306436538696e-08
 33. $\frac{1}{8589934592}$ of 100 = 1.164153218269348e-08
 34. $\frac{1}{17179869184}$ of 100 = 5.82076609134674e-09
 35. $\frac{1}{34359738368}$ of 100 = 2.91038304567337e-09
 36. $\frac{1}{68719476736}$ of 100 = 1.455191522836685e-09
 37. $\frac{1}{137438953472}$ of 100 = 7.275957614183425e-10
 38. $\frac{1}{274877906944}$ of 100 = 3.6379788070917125e-10
 39. $\frac{1}{549755813888}$ of 100 = 1.8189894035458562e-10
 40. $\frac{1}{1099511627776}$ of 100 = 9.094947017729281e-11
 41. $\frac{1}{2199023255552}$ of 100 = 4.5474735088646405e-11
 42. $\frac{1}{4398046511104}$ of 100 = 2.2737367544323202e-11
 43. $\frac{1}{8796093022208}$ of 100 = 1.1368683772161601e-11
 44. $\frac{1}{17592186044416}$ of 100 = 5.6843418860808005e-12
 45. $\frac{1}{35184372088832}$ of 100 = 2.8421709430404002e-12
 46. $\frac{1}{70368744177664}$ of 100 = 1.4210854715202001e-12
 47. $\frac{1}{140737488355328}$ of 100 = 7.1054273576010005e-13
 48. $\frac{1}{281474976710656}$ of 100 = 3.5527136788005002e-13
 49. $\frac{1}{562949953421312}$ of 100 = 1.7763568394002501e-13
 50. $\frac{1}{1125899906842624}$ of 100 = 8.8817841970012505e-14
 51. $\frac{1}{2251799813685248}$ of 100 = 4.4408920985006252e-14
 52. $\frac{1}{4503599627370496}$ of 100 = 2.2204460492503126e-14
 53. $\frac{1}{9007199254740992}$ of 100 = 1.1102230246251563e-14
 54. $\frac{1}{18014398509481984}$ of 100 = 5.5511151231257815e-15
 55. $\frac{1}{36028797018963968}$ of 100 = 2.7755575615628907e-15
 56. $\frac{1}{72057594037927936}$ of 100 = 1.3877787807814454e-15
 57. $\frac{1}{144115188075855872}$ of 100 = 6.938893903907227e-16
 58. $\frac{1}{288230376151711744}$ of 100 = 3.4694469519536135e-16
 59. $\frac{1}{576460752303423488}$ of 100 = 1.7347234759768068e-16
 60. $\frac{1}{1152921504606846976}$ of 100 = 8.673617379884034e-17
 61. $\frac{1}{2305843009213693952}$ of 100 = 4.336808689942017e-17
 62. $\frac{1}{4611686018427387904}$ of 100 = 2.1684043449710085e-17
 63. $\frac{1}{9223372036854775808}$ of 100 = 1.0842021724855042e-17
 64. $\frac{1}{18446744073709551616}$ of 100 = 5.421010862427521e-18
 65. $\frac{1}{36893488147419103232}$ of 100 = 2.7105054312137605e-18
 66. $\frac{1}{73786976294838206464}$ of 100 = 1.3552527156068802e-18
 67. $\frac{1}{147573952589676412928}$ of 100 = 6.776263578034401e-19
 68. $\frac{1}{295147905179352825856}$ of 100 = 3.3881317890172005e-19
 69. $\frac{1}{590295810358705651712}$ of 100 = 1.6940658945086002e-19
 70. $\frac{1}{1180591620717411303424}$ of 100 = 8.470329472543001e-20
 71. $\frac{1}{2361183241434822606848}$ of 100 = 4.2351647362715005e-20
 72. $\frac{1}{4722366482869645213696}$ of 100 = 2.1175823681357502e-20
 73. $\frac{1}{9444732965739290427392}$ of 100 = 1.0587911840678751e-20
 74. $\frac{1}{18889465931478580854784}$ of 100 = 5.2939559203393755e-21